Visual Complexity (VC) Prediction Pipeline

Automated VC scoring of 520 visualization images via Claude Opus API with few-shot anchored prompting.

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1. Method

Overview

Each image is scored by Claude Opus (`claude-opus-4-6`) on 7 VC dimensions (0–1) plus an overall `vc_score`. The model receives a system prompt defining dimensions and calibration guidance, followed by 3 few-shot anchor image–response pairs, then the target image. The prompt was iteratively refined across 3 versions using 46 ground-truth images.

Few-Shot Anchor Images

Three hand-scored anchor images are injected as conversation-turn examples (user image → assistant JSON) spanning the low–mid–high VC range:

Anchor Image	VC Score	Description
`VisC.503.6.png`	0.22	Simple 3-bar chart with black hatching and error bars.
`InfoVisJ.619.17.png`	0.54	PCA biplot with directional arrows and grouped points.
`InfoVisJ.1149.6(1).png`	0.95	Multi-panel text collation tool with 7 coordinated views.

API Message Structure

```
┌──────────────────────────────────────────────┐
│ system: SYSTEM_PROMPT                        │  ← cache_control: ephemeral
├──────────────────────────────────────────────┤
│ user:   [anchor image 1]                     │
│         "Score the visual complexity of this..."│
│ asst:   { "data_density": 0.15, ... }        │  Anchor 1 (VC=0.22)
├──────────────────────────────────────────────┤
│ user:   [anchor image 2]                     │
│         "Score the visual complexity of this..."│
│ asst:   { "data_density": 0.55, ... }        │  Anchor 2 (VC=0.54)
├──────────────────────────────────────────────┤
│ user:   [anchor image 3]                     │
│         "Score the visual complexity of this..."│
│ asst:   { "data_density": 0.90, ... }        │  Anchor 3 (VC=0.95)
│                                              │  ← cache_control: ephemeral
├──────────────────────────────────────────────┤
│ user:   [TARGET IMAGE]                       │  ← only this varies per call
│         "Score the visual complexity of this..."│
└──────────────────────────────────────────────┘
```

The system prompt + 3 anchor turns are **identical across all 520 calls** and cached after the first request (see Appendix B). Only the final user message changes per call.

Prompt Evolution (V1 → V2 → V3)

Metric	V1	V2	V3
Pearson r	0.832	0.885	**0.912**
Spearman ρ	0.818	0.873	**0.907**
MAE	0.097	0.083	**0.074**
RMSE	0.120	0.100	**0.089**
Bias	−0.085	−0.054	**−0.025**

Key V3 changes: "push scores up" instruction, `text_annotation` scored on volume only, explicit dimension weighting for `vc_score` (high weight on `data_density`, `visual_encoding`, `domain_schema`, `cognitive_load`; low weight on `text_annotation`).

Production Run Configuration

Parameter	Value
Model	`claude-opus-4-6`
Total images	520 (510 after VisType filtering)
Image source	GitHub URLs (`AllDataResize/`)
Max tokens	1,500
Concurrency	Async with semaphore-bounded workers

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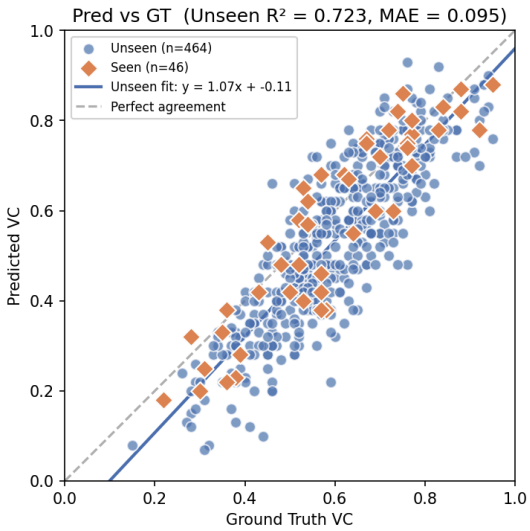
2. Ground Truth Validation

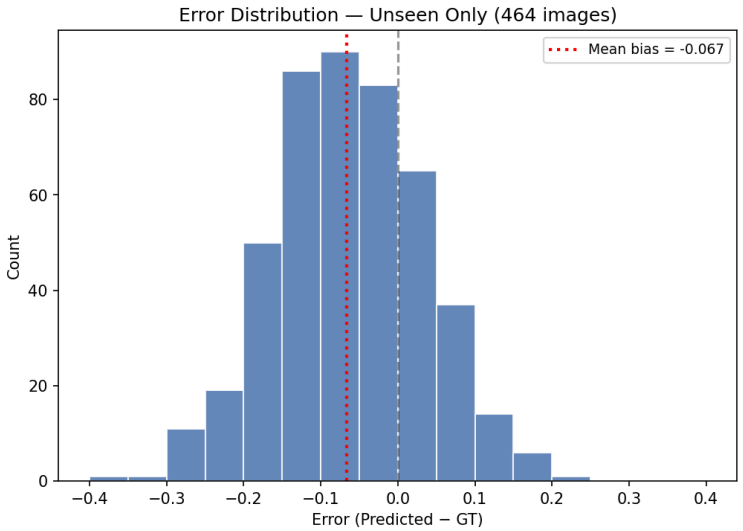
The V3 prompt was developed using 3 anchor images + 46 calibration images = **49 "seen" images**. To report unbiased prediction quality, we split into:

- **Seen** (49 images): anchors + calibration set used during prompt development.
- **Unseen** (~461 images): purely predicted, never used for prompt tuning.

Split	Pearson r	Spearman ρ	R^2	MAE	RMSE	Bias
All (n=510)	0.731	0.729	0.534	0.107	0.135	−0.020
Seen (n=49)	0.912	0.907	0.832	0.074	0.089	−0.025
Unseen (n=461)	0.718	0.716	0.516	0.111	0.138	−0.019

(Exact values from notebook — re-run to confirm.)





The unseen split shows Pearson r = 0.718 with minimal bias (−0.019), indicating the prompt generalizes beyond the calibration set. The error distribution is approximately symmetric and centered near zero.

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3. File Reference

File	Description
`scripts/_vc_score_api.py`	Main API scoring script (V3 prompt, async, caching)
`vc_api_scores/vc_scores.csv`	Production scores (520 rows, 7 dims + vc_score)
`vc_api_scores/vc_explanations.csv`	Per-dimension explanations + summary
`Claude_image_pair_comparison/gt_all_46.csv`	46 calibration GT images
`Claude_image_pair_comparison/VC_520_Analysis.ipynb`	Production analysis notebook
`Claude_image_pair_comparison/VC_API_Calibration.ipynb`	V1/V2/V3 calibration diagnostics
`vc_api_46gt/`	V1 calibration run output
`vc_api_46gt_v2/`	V2 calibration run output
`vc_api_46gt_v3/`	V3 calibration run output

Appendices

Appendix A: Full V3 System Prompt

```
You are a visual complexity (VC) scoring expert for data visualizations.

**Visual Complexity** measures the amount of detail, intricacy, and
```

```
perceptual/cognitive
demand of a visualization image. It is scored on a 0–1 scale where 0 = minimal
complexity
and 1 = maximum complexity.

You will receive a single visualization image with NO accompanying text or
comments.
Score it purely from what you see.

## Calibration Guidance
Use the full 0–1 range. Do NOT cluster scores conservatively toward the center.
- A plain single bar chart or pie chart with < 5 elements → vc_score ≈ 0.25–0.40
- A standard scatterplot, grouped bar chart, or line chart → vc_score ≈ 0.45–0.60
- A multi-panel dashboard, dense network, or multi-encoding composite → vc_score ≈
0.70–0.85
- An extremely dense multi-view system with many coordinated panels → vc_score ≈
0.85–0.95
IMPORTANT: If you are unsure between two scores, always choose the higher one.
Most scorers underestimate — push scores up.

## Scoring Dimensions (each 0–1)

1. **data_density** — The perceived amount, richness, or depth of data content.
   Considers information volume, element quantity, and visual clutter/overlap.
2. **visual_encoding** — The variety, type, and complexity of graphical forms
   (shapes, lines, marks) and how spatial layout, scale, and encoding
   interpretability contribute to complexity.
3. **text_annotation** — The sheer quantity and density of text elements
   (titles, axis labels, legends, captions, annotations, in-chart labels).
   Score based on volume only: more text = higher score, regardless of legibility.
4. **domain_schema** — Whether specialized domain knowledge is needed,
   including dimensionality (2D/3D), structural complexity, and abstraction level.
5. **color_symbol** — Range, variety, and arrangement of colors, plus use of
   symbols, textures, and non-color graphical markers.
6. **aesthetic_order** — How visually cluttered, dense, or disordered the layout
   appears. Higher = more cluttered/overwhelming. A clean minimal layout scores
   low; a crowded layout with overlapping elements scores high.
7. **cognitive_load** — Overall ease or difficulty of interpreting the
   visualization. Considers interpretive difficulty, semantic clarity, and
   processing time/effort.

## Overall vc_score
The overall vc_score is a weighted average reflecting the image's holistic visual
complexity. Weight dimensions as follows:
- **High weight**: data_density, visual_encoding, domain_schema, cognitive_load
- **Medium weight**: color_symbol, aesthetic_order
- **Low weight**: text_annotation (text volume alone is a weak VC signal)
The vc_score should be consistent with (but not necessarily the arithmetic mean
of)
the 7 dimension scores. A low text_annotation score should NOT substantially pull
down the overall vc_score.

## Output Format
Return ONLY valid JSON (no markdown fences, no explanation outside JSON):
```

```
{
  "data_density": <float 0-1>,
  "visual_encoding": <float 0-1>,
  "text_annotation": <float 0-1>,
  "domain_schema": <float 0-1>,
  "color_symbol": <float 0-1>,
  "aesthetic_order": <float 0-1>,
  "cognitive_load": <float 0-1>,
  "vc_score": <float 0-1>,
  "data_density_explanation": "<1 sentence>",
  "visual_encoding_explanation": "<1 sentence>",
  "text_annotation_explanation": "<1 sentence>",
  "domain_schema_explanation": "<1 sentence>",
  "color_symbol_explanation": "<1 sentence>",
  "aesthetic_order_explanation": "<1 sentence>",
  "cognitive_load_explanation": "<1 sentence>",
  "summary": "<2-3 sentence overall description>"
}
```

Appendix B: Prompt Caching

The Anthropic API supports **prompt caching** to avoid re-sending identical prefix content on every call. Since all 520 requests share the same system prompt and 3 anchor image/response pairs, caching reduces token costs significantly.

How It Works

The **system prompt** is marked with `cache_control: {"type": "ephemeral"}`:

```
system=[{"type": "text", "text": SYSTEM_PROMPT, "cache_control": {"type":
"ephemeral"}}]
```

The **last anchor assistant message** (3rd few-shot response) is also marked:

```
last_msg['content'] = [{
    "type": "text",
    "text": last_msg['content'],
    "cache_control": {"type": "ephemeral"}
}]
```

Why Prefix-Based Caching

Anthropic's caching is **prefix-based** — it caches everything from the start up to the last `cache_control` block. The two markers serve complementary roles:

- **System prompt marker**: ensures the system text is cached even if anchor image loading fails.

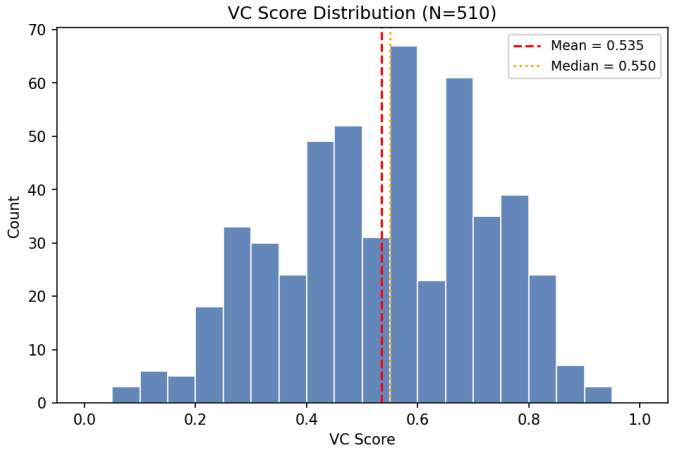
- **Last anchor marker**: extends the cached prefix to include all 3 few-shot examples.

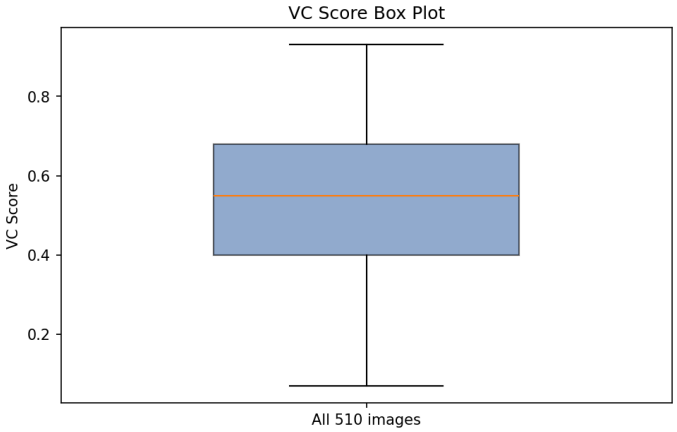
On the 2nd call onward, the API performs a **cache read** for the full prefix (~5,000+ tokens) and only processes the new target image. Cache reads cost 10% of normal input token pricing.

Appendix C: Distribution & Dimension Analysis

Overall VC Score Distribution

- **Mean ± std**: 0.531 ± 0.168
- **Range**: [0.03, 0.93]



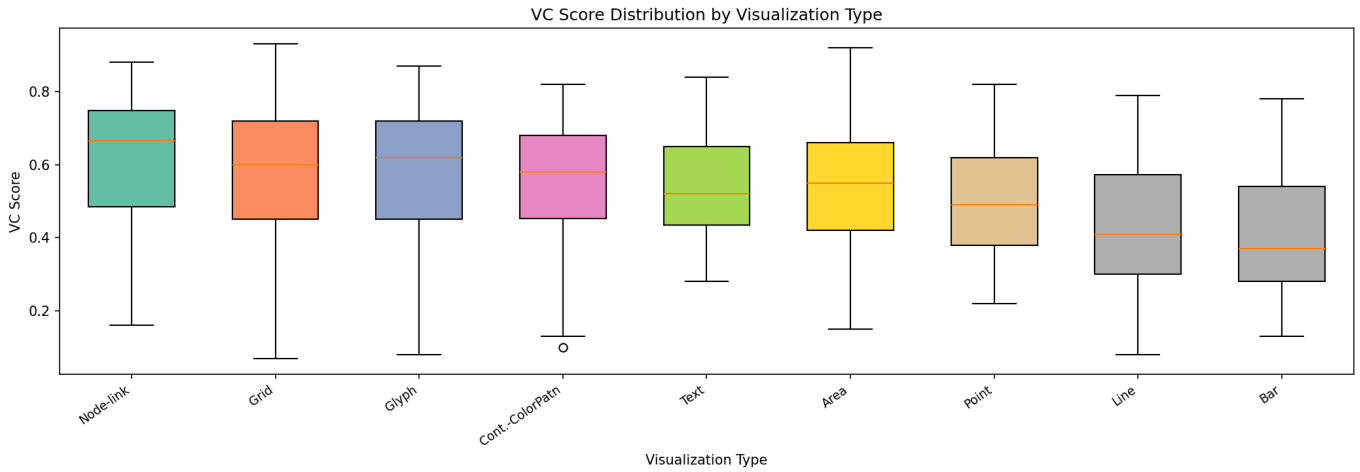


VC by Visualization Type

9 core visualization types retained (510 images):

VisType	Count
Node-link	66
Area	65
Grid	65
Glyph	64
Point	58
Bar	51
Text	51
Line	48
Cont.-ColorPatn	42

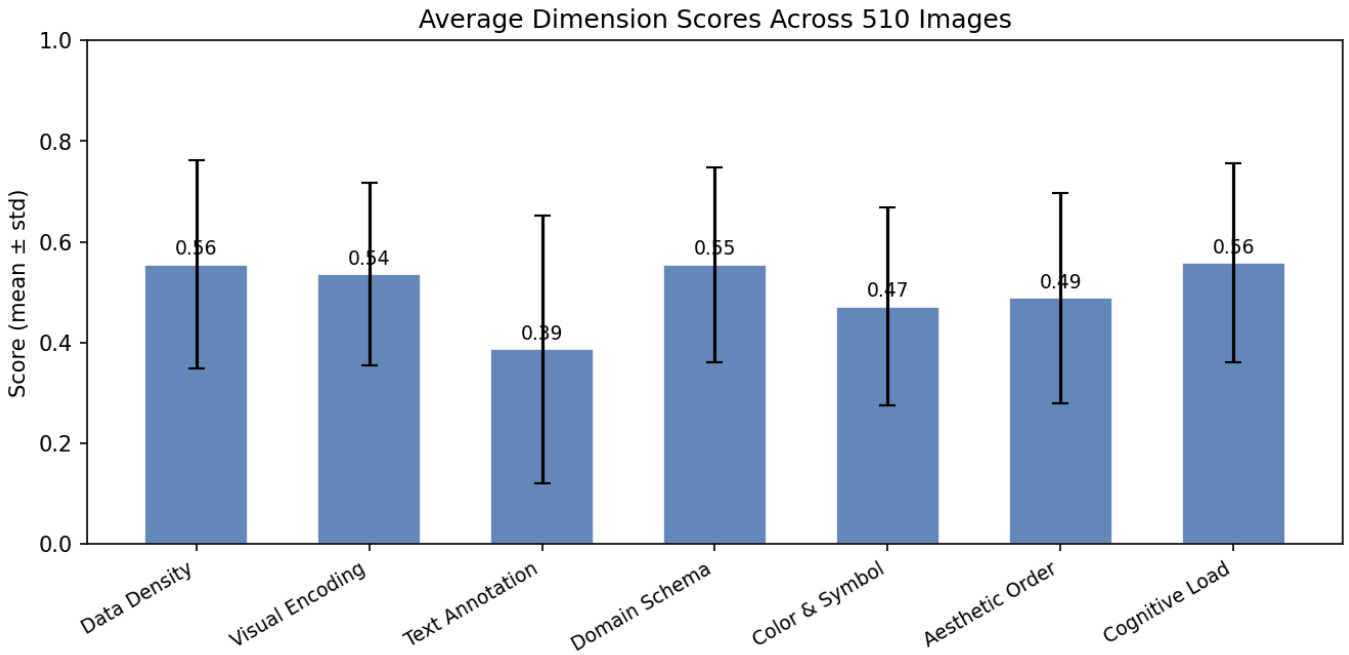
Excluded (10 images): Schematic (7), Bar and point (1), Table (1), Area and Text (1).

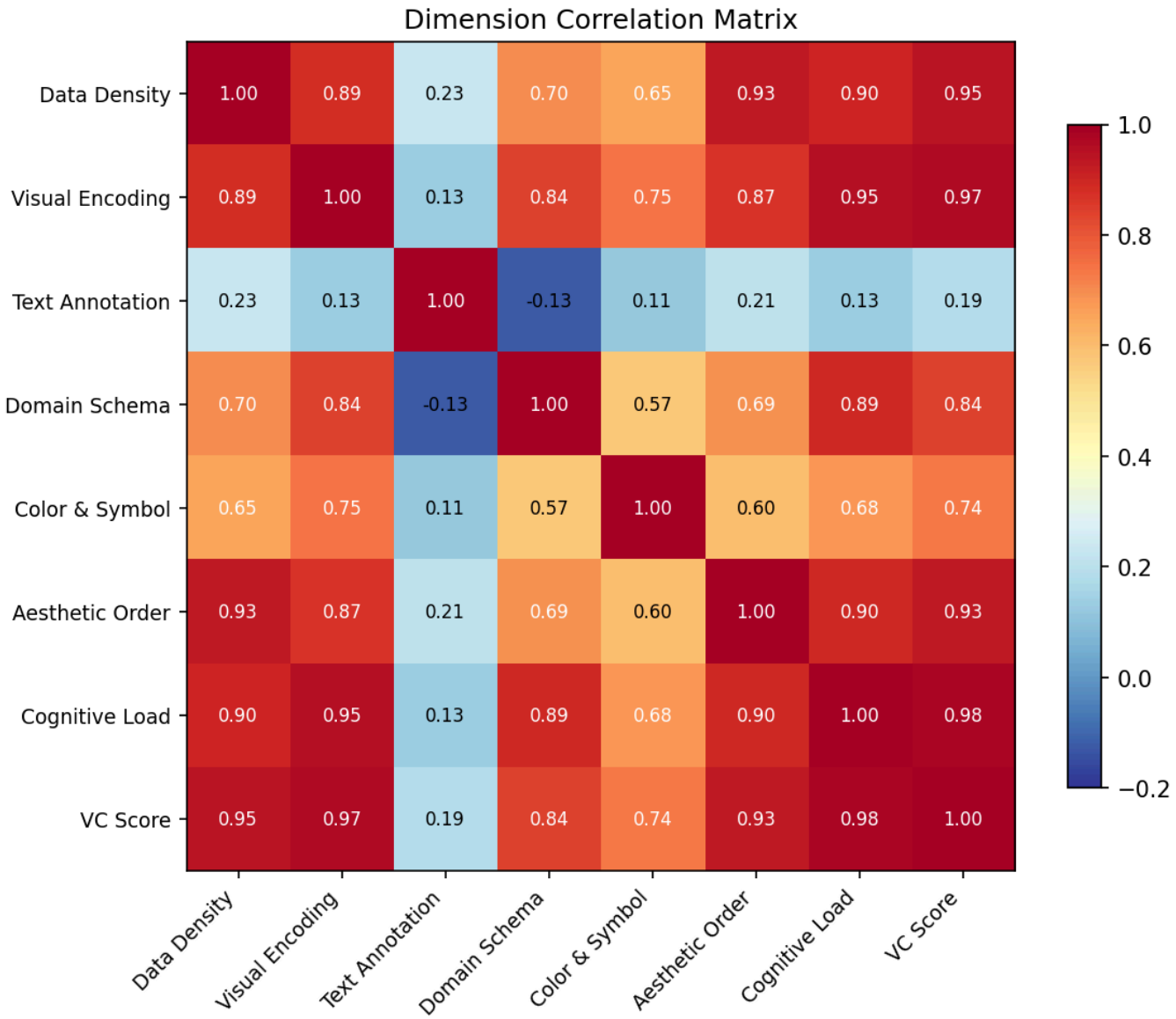


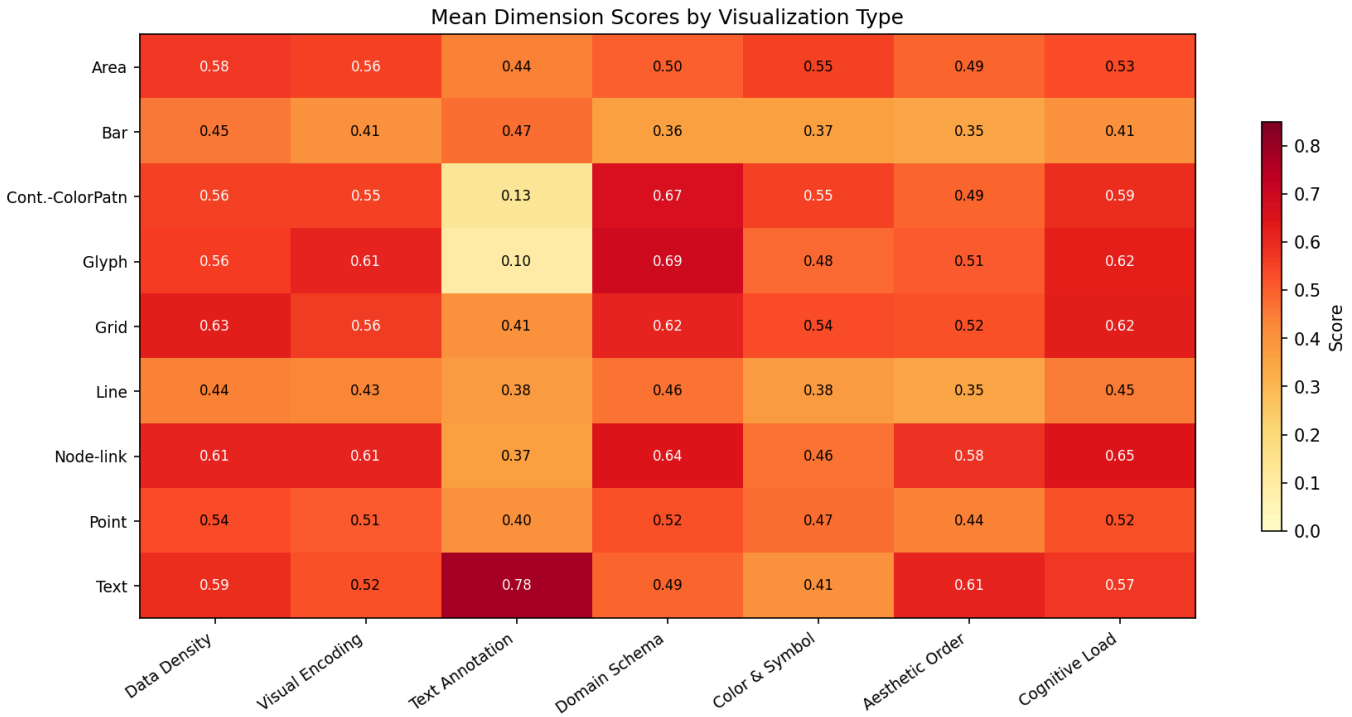
Dimension-Level Statistics

Dimension	Mean	Std
Data Density	0.56	0.19
Visual Encoding	0.55	0.17
Text Annotation	0.47	0.17
Domain Schema	0.55	0.19
Color & Symbol	0.48	0.19
Aesthetic Order	0.52	0.18
Cognitive Load	0.57	0.18

(Approximate values — see notebook for exact stats.)




Dimension Correlation Matrix
Data Density 1.00 0.89 0.23 0.70 0.65 0.93 0.90 0.95
Visual Encoding 0.89 1.00 0.13 0.84 0.75 0.87 0.95 0.97
Text Annotation 0.23 0.13 1.00 -0.13 0.11 0.21 0.13 0.19
Domain Schema 0.70 0.84 -0.13 1.00 0.57 0.69 0.89 0.84
Color & Symbol 0.65 0.75 0.11 0.57 1.00 0.60 0.68 0.74
Aesthetic Order 0.93 0.87 0.21 0.69 0.60 1.00 0.90 0.93
Cognitive Load 0.90 0.95 0.13 0.89 0.68 0.90 1.00 0.98
VC Score 0.95 0.97 0.19 0.84 0.74 0.93 0.98 1.00
Data Density
Visual Encoding
Text Annotation
Domain Schema
Color & Symbol
Aesthetic Order
Cognitive Load
VC Score
1.0
0.8
0.6
0.4
0.2
0.0
−0.2


Mean Dimension Scores by Visualization Type
Area 0.58 0.56 0.44 0.50 0.55 0.49 0.53
Bar 0.45 0.41 0.47 0.36 0.37 0.35 0.41
Cont.-ColorPatn 0.56 0.55 0.13 0.67 0.55 0.49 0.59
Glyph 0.56 0.61 0.10 0.69 0.48 0.51 0.62
Grid 0.63 0.56 0.41 0.62 0.54 0.52 0.62
Line 0.44 0.43 0.38 0.46 0.38 0.35 0.45
Node-link 0.61 0.61 0.37 0.64 0.46 0.58 0.65
Point 0.54 0.51 0.40 0.52 0.47 0.44 0.52
Text 0.59 0.52 0.78 0.49 0.41 0.61 0.57
Data Density
Visual Encoding
Text Annotation
Domain Schema
Color & Symbol
Aesthetic Order
Cognitive Load
Score
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

Appendix D: Why Conversation-Turn Anchoring

A natural question is whether the 3 few-shot anchors could be combined into a single text message (e.g., "Here are 3 reference images with their scores: ...") rather than using separate user→assistant conversation turns.

Why separate turns are preferred

1. **Images require user messages.** The Claude API only accepts images inside user-role messages. Placing all 3 anchor images in a single user message would make it ambiguous which image corresponds to which JSON response.

2. **Behavioral anchoring through synthetic history.** When the model sees that it "already answered" correctly 3 times in prior assistant turns, it follows the same scoring pattern much more reliably than if the examples are described in prose. The alternating user→assistant turn structure is the canonical few-shot format for chat-based LLM APIs — it leverages the model's in-context learning from its own prior responses.

3. **Exact output format teaching.** Each assistant turn contains the full JSON output with all 7 dimension scores, explanations, and summary. The model learns the precise structure by having "produced" it 3 times, rather than merely reading a format specification.

The alternative (and why it's weaker)

You *could* describe the anchors in text within the system prompt: "Image A is a simple bar chart, VC = 0.22..." This would work for non-visual tasks, but here the model would be scoring based on text descriptions rather than seeing the actual images. The conversation-turn approach ensures the model has seen real visual examples at each complexity level before scoring the target.